

Subject-Aware Diffusion Models for Cross-Subject EEG Denoising

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Electroencephalography (EEG) is widely used in cognitive neuroscience, clinical monitoring, and brain-computer interfaces (BCIs), but its low signal-to-noise ratio and strong inter-subject variability make denoising hard to generalize across users. We propose a Subject-Aware Denoising Diffusion Probabilistic Model (SADDPM): on top of the standard DDPM forward/reverse mechanism, it introduces learnable subject-embedding vectors injected at multiple network levels, so the reverse denoising trajectory adapts to each individual. On BCI Competition IV-2a, downstream motor-imagery classification and a subject-consistency analysis show SADDPM is more stable than traditional baselines across source-target pairs, with gains concentrated in cross-subject transfer and individual structure preserved. To assess fidelity directly, a conditional paired-data variant, SADDPM-Cond, is competitive with supervised denoisers on single-channel ocular-artifact removal, achieves high-fidelity joint multi-channel denoising, and shows functionally relevant conditioning when artifacts are subject-specific. These results support subject-aware diffusion as a viable adaptive framework for EEG denoising and BCI preprocessing.

CCS Concepts: • **Computing methodologies** → **Neural networks**; • **Applied computing** → *Health informatics*.

Additional Key Words and Phrases: EEG denoising, denoising diffusion probabilistic model, subject-aware learning, cross-subject generalization, conditional generative model, brain-computer interface

ACM Reference Format:

Tao HE, Zhilin LI, and Yinghao WANG. 2026. Subject-Aware Diffusion Models for Cross-Subject EEG Denoising. *ACM Trans. Auton. Adapt. Syst.* 0, 0, Article 0 (January 2026), 30 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Electroencephalography (EEG) is a physiological signal that records brain neural activity non-invasively, and it is widely used in cognitive neuroscience, clinical neurology, and brain-computer interfaces (BCIs) [6]. Owing to its high temporal resolution and convenient acquisition, EEG plays a key role in neural-function monitoring, computer-aided diagnosis, and assessment of states of consciousness. Nevertheless, EEG has an intrinsically low signal-to-noise ratio and is easily contaminated by a variety of artifacts and environmental noise, including ocular (EOG) and electromyographic (EMG) activity, power-line interference (e.g., the 50/60 Hz line frequency), and poor electrode contact. Such contamination severely distorts the underlying neural information and degrades the accuracy of downstream analyses [28, 31].

To suppress noise and extract reliable neural features, EEG denoising has long been a core problem in the field. Traditional approaches—most prominently band-pass filtering and Independent Component Analysis (ICA)—perform well in specific settings but suffer from two key bottlenecks. First, they are highly sensitive to parameter choices and to assumptions about the noise distribution, which makes them difficult to adapt to the dynamic and complex conditions of real-world acquisition [11]. Second, they largely ignore inter-individual differences in cranial anatomy, cognitive state, and acquisition conditions, so their generalization deteriorates in cross-subject scenarios [16].

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In recent years, deep learning has provided new tools for modeling and denoising EEG signals. Following the success of generative adversarial networks (GANs) and variational autoencoders (VAEs) in image denoising and audio reconstruction, considerable effort has been devoted to applying such models to neural-signal processing [29]. However, these models typically require large amounts of identically distributed training data; faced with the high heterogeneity and non-stationarity of EEG, they still struggle with generalization and training stability.

Denoising Diffusion Probabilistic Models (DDPMs) are a recently emerged generative framework that has achieved major breakthroughs in image and audio generation [14], thanks to progressive reconstruction, stable training, and a strong ability to model complex distributions. The core idea is to gradually perturb data into noise and then recover clean samples through a reverse diffusion process. This layer-by-layer formulation is naturally aligned with the need to progressively purify EEG signals and restore their structure. Although diffusion models have been explored for the reconstruction of other physiological signals such as the electrocardiogram [1, 12], DDPM-based modeling that accounts for cross-subject individual differences in EEG remains at an early stage and has not been systematically studied [30].

This paper presents a new strategy for cross-subject EEG denoising, the *Subject-Aware Denoising Diffusion Probabilistic Model* (SADDPM). To the best of our knowledge, this is among the first attempts to incorporate a subject-aware mechanism into the DDPM denoising pipeline, endowing the model with the ability to recognize and adapt to inter-individual domain differences. During modeling, a distinct embedding vector is learned for each subject and is injected as a conditioning signal at multiple layers of the reverse process, so that the model can dynamically adjust its noise-estimation trajectory according to the characteristics of each subject, so as to pursue more robust and transferable denoising. The central idea is to combine the expressive power of generative modeling with the generalization afforded by an individual-aware mechanism, yielding a denoising model that is physiologically motivated and has practical potential for handling the structural artifacts and individual variability inherent in EEG. In short, this work builds a DDPM-based, domain-specific EEG denoising method that explicitly targets the weak cross-subject generalization of existing approaches. From the perspective of autonomous and adaptive systems, an EEG pipeline that must remain effective for new users is itself a learning-based system facing distribution shift, and our subject-aware diffusion model instantiates the emerging use of generative AI as a mechanism for such adaptation [5, 17]. Our main contributions are summarized as follows.

- We design and implement a domain-specific denoising diffusion probabilistic model (SADDPM) that exploits the step-by-step denoising of diffusion models to effectively distinguish the noise and signal characteristics of different subjects.
- Through a transductive, label-free target-subject adaptation strategy, SADDPM estimates a deployment subject's embedding from unlabeled target EEG and adapts the reverse denoising trajectory without using target task labels, clean targets, or artifact annotations.
- We construct a complete pipeline—covering data processing, model implementation, and evaluation—and conduct systematic experiments on the BCI Competition IV-2a dataset, validating the effectiveness of the proposed method under complex noise conditions and in cross-subject tasks.
- We complement the downstream-classification study with a paired ground-truth denoising evaluation under the EEGdenoiseNet protocol using a conditional variant, SADDPM-Cond. The results show competitive single-channel ocular-artifact removal, high-fidelity multi-channel denoising, and functionally relevant subject conditioning when artifacts are subject-specific.

The remainder of this paper is organized as follows. Section 2 reviews related work on EEG denoising, subject variability, and diffusion-based signal modeling. Section 3 details the proposed SADDPM, including its diffusion formulation, subject-aware conditioning, network architecture, and training objective. Section 4 describes the dataset, preprocessing, experimental protocol, and downstream results; Section 5 reports the paired ground-truth denoising study. Section 6 concludes the paper and discusses future directions.

2 Related Work

2.1 EEG Denoising

2.1.1 Traditional EEG Denoising Methods. Before the rise of deep learning, classical signal-processing techniques dominated EEG denoising for a long time. Filtering, the wavelet transform, and ICA, in particular, were widely adopted in practical EEG preprocessing because they are theoretically mature and simple to implement. Each can effectively improve EEG quality under suitable conditions, yet each also has inherent limitations when confronted with complex, non-stationary, multi-source noise.

Filtering is the most basic denoising approach: it removes noise components in specific frequency bands—such as low-frequency baseline drift (< 0.5 Hz) and high-frequency EMG interference (> 40 Hz)—using band-pass or notch filters [31]. These methods are computationally cheap and suitable for real-time use, but they assume that signal and noise are strictly separable in frequency. This assumption fails when the spectra overlap; for example, ocular artifacts (EOG) overlap with the δ band (0.5–4 Hz). Moreover, an inappropriate filter design can introduce phase distortion or remove useful information, harming subsequent analysis.

The wavelet transform, owing to its good time–frequency localization, is another common tool. Wavelet denoising decomposes the signal into sub-bands at multiple scales and applies thresholding to high-frequency coefficients. Unlike conventional filtering, it adapts to signal non-stationarity and is well suited to transient artifacts such as the EOG produced by an eye blink. However, its performance depends strongly on the choice of wavelet basis, the number of decomposition levels, and the thresholding strategy, and it struggles with strongly non-additive or multi-source interference (e.g., EMG superimposed on ambient electromagnetic noise). Over-thresholding may also remove useful neural components and damage the original signal structure.

ICA, a blind source separation (BSS) technique, is widely used for artifact removal. It assumes that the observed EEG is a linear mixture of statistically independent sources and separates neural signals from ocular and muscular artifacts by maximizing the independence of the recovered components. ICA is particularly suited to multi-channel EEG and can extract the dominant artifact components in an unsupervised manner. Its limitations, however, are pronounced: (i) the independence assumption does not always hold for real EEG; (ii) identifying artifact components typically requires manual intervention or auxiliary criteria, increasing application complexity; and (iii) ICA is sensitive to the number of electrodes and the amount of data, with markedly reduced separation quality when channels are few or the noise level is high [16].

In summary, filtering, the wavelet transform, and ICA can improve EEG quality under specific conditions, but they generally rely on prior assumptions about the noise (e.g., frequency distribution, independence), lack adaptivity, and cannot automatically accommodate the differing signal characteristics of different subjects. They therefore exhibit a serious lack of cross-subject generalization, motivating a shift toward data-driven methods with stronger modeling capacity [7, 19].

2.1.2 Deep-Learning-Based Denoising Methods. Deep learning has attracted considerable attention for EEG denoising. Models such as convolutional neural networks (CNNs), recurrent neural

networks (RNNs), and denoising autoencoders (DAEs) recover EEG under complex noise through end-to-end feature extraction and reconstruction.

CNNs are widely adopted because of their strong local-feature-extraction ability. Common designs model the time-domain signal with one-dimensional convolutions (1D-CNN) or extract features from time-frequency representations (e.g., the short-time Fourier transform or wavelet transform) with two-dimensional convolutions (2D-CNN) [31]. CNNs perform well at suppressing high-frequency EMG and EOG artifacts while preserving the local structure of the waveform. However, the limited receptive field of convolutions focuses on local features and captures long-range temporal dependencies poorly, with limited ability to handle non-stationarity and cross-scale dynamics.

RNNs and their variants—long short-term memory (LSTM) and gated recurrent units (GRU)—are used to strengthen temporal modeling. Their recurrent structure captures long-range temporal dependencies and suits artifacts with temporal correlation, such as EOG and baseline drift. Some studies further introduce attention mechanisms to emphasize salient segments [6], or combine CNNs and RNNs so that front-end convolutions extract local features while a back-end recurrent module captures global temporal dependencies. Despite their strength in modeling complex temporal dynamics, RNNs are prone to vanishing or exploding gradients because of their sequential processing, and their inference is relatively slow, increasing computational cost in practice.

DAEs, an unsupervised framework, are widely used for EEG noise suppression. During training, a DAE adds noise to the input and learns to reconstruct the clean signal through an encoder-decoder, thereby capturing the structural features of the signal. Beyond the standard shallow architecture, convolutional DAEs (CDAEs) and stacked DAEs (SDAEs) have been developed to improve nonlinear feature extraction and denoising under complex noise. DAEs are relatively stable on small datasets, adapt well to weakly labeled or unlabeled data, and are simple to train [8]. However, most DAEs adopt the mean squared error (MSE) as the reconstruction loss, which struggles to model non-Gaussian, non-additive artifacts such as EOG and EMG [11], and they lack an explicit characterization of the temporal dynamics of EEG, reducing denoising performance across time segments and across subjects.

Overall, CNN-, RNN-, and DAE-based methods achieve high recovery quality for specific noise types and handle short-duration, high-frequency noise and local bursts well, but they remain limited in modeling capacity and adaptivity when the signal distribution changes drastically, the noise is complex and multi-source, or strong cross-subject generalization is required. How to combine multi-scale feature extraction, temporal modeling, and domain-adaptation strategies to improve robustness in complex environments is therefore a key direction for further research.

2.2 Individual Variability in EEG

Individual variability in EEG arises from differences in physiology and acquisition conditions. Anatomical factors—such as cortical folding patterns, skull thickness, and scalp tissue structure—directly affect the attenuation and spread of neural activity during volume conduction, causing the amplitude and spatial distribution of scalp EEG to differ across subjects [16]. Physiological and cognitive factors—age, sex, health status (e.g., fatigue, medication, neurological disease), and psychological state (e.g., attention, mood)—further alter the spectral composition and temporal dynamics of the signal [28]. Minor differences in acquisition conditions, such as electrode placement, contact impedance, and hardware noise level, add to the inconsistency of the data distribution across subjects [11].

Because of these differences, even under identical tasks and external conditions, the EEG of different subjects exhibits a systematic *distribution shift*. This shift directly affects data-driven modeling: a model trained on a single subject often degrades markedly when transferred to a new

subject, which is a key bottleneck for the large-scale deployment of EEG analysis systems. Viewed through the lens of autonomous and adaptive systems, this cross-subject distribution shift is a form of *adaptation-space drift*: the conditions under which a learning-based system operates change across deployment contexts, so the model must track and adapt to them rather than assume a fixed data distribution [10].

To address this, *subject-aware* modeling has recently been introduced to characterize and adapt to the individual variability that pervades EEG data [16]. The premise is that each subject has specific signal characteristics, artifact-contamination patterns, and background conditions; for example, some subjects show power-spectral densities skewed toward low frequencies while others show more prominent high-frequency components [28]. Blink rate, muscle-activity level, and baseline-drift amplitude also differ systematically across subjects. By explicitly injecting individual-specific information into the denoising and feature-modeling process, subject-aware modeling helps a model adapt its signal recovery and feature extraction, in principle overcoming the limitations of a single subject-generalized model and improving cross-subject adaptivity and stability.

In EEG denoising, the distribution differences between subjects give rise to a *subject gap*: even when good performance is achieved on a source subject, the model often degrades substantially when transferred to a target subject [6]. This is caused by changes in the underlying EEG characteristics, differences in artifact-contamination patterns (e.g., EMG intensity, EOG frequency), and varying background signal-to-noise ratios. Designing denoising methods that perceive subject characteristics and adapt to individual differences is therefore central to improving the cross-subject robustness and practical usability of EEG systems [15].

2.3 Diffusion Probabilistic Models for Signal Denoising

Compared with traditional EEG denoising methods such as band-pass filtering and ICA, DDPM-based approaches offer several advantages under complex noise. First, DDPMs enjoy stable training: their loss is defined directly on noise prediction and requires no discriminator, avoiding the mode collapse and training instability that commonly affect GANs [14] and improving controllability and convergence. Second, DDPMs reconstruct signals progressively, capturing both local and global features step by step, preserving fine details and suppressing the information loss that denoising can introduce [1]. Third, diffusion models can in principle approximate arbitrarily complex data distributions, flexibly modeling the highly non-stationary and heterogeneous multi-source contamination of EEG [30]. Finally, the DDPM framework naturally supports conditional generation, allowing auxiliary information such as a subject embedding to be incorporated for subject-aware denoising [9], thereby improving generalization and recovery quality in cross-subject settings.

By bringing the DDPM framework to EEG denoising and combining it with subject-aware modeling, the proposed SADDPM aims to improve the stability and robustness of denoising under complex noise and high inter-subject heterogeneity [30]. In contrast, most current deep models (e.g., CNNs, RNNs, and DAEs) focus on local features or short-term temporal dynamics and lack a systematic mechanism to adapt to inter-subject domain specificity; they are typically trained and validated on single-subject or homogeneous datasets, without fully addressing cross-subject transfer and domain adaptation, and they exhibit large performance fluctuations in open, real-world environments.

While generative models such as GANs have shown potential for EEG generation and denoising [2, 13, 18], their training instability, mode collapse, and evaluation difficulties limit their use in reliability-critical applications. DDPMs, by contrast, have achieved breakthrough results in image generation and reconstruction through stable training, progressive recovery, and high-fidelity reconstruction [3, 14, 20, 21, 25]. Yet research on DDPM-based EEG denoising remains largely

unexplored: existing diffusion applications concentrate on natural signals such as images and audio, with little systematic study of EEG data, which are multi-source, heterogeneous, and highly non-stationary. The progressive-reconstruction and conditional-control properties of diffusion models make them theoretically well suited to building a noise-removal process for complex EEG, with the potential—through domain-specific information such as subject labels or feature encodings—to achieve adaptive denoising across subjects and heterogeneous domains.

Beyond signal processing, generative AI is increasingly viewed as a mechanism for adaptation in autonomous and adaptive systems [17], including the use of generative models to empower human-activity recognition in autonomic systems [27]; relatedly, deep learning has been used to manage large adaptation spaces in self-adaptive systems [26]. Our subject-aware diffusion model fits this trend, using a conditional generative model to adapt the denoising process to each individual domain. These observations motivate the domain-specific, subject-aware diffusion approach developed in this paper.

3 Method: Subject-Aware Denoising Diffusion Probabilistic Model

We propose the Subject-Aware Denoising Diffusion Probabilistic Model (SADDPM), which introduces subject-level adaptation into the learned reverse process of a diffusion model for multi-subject EEG denoising. Importantly, SADDPM keeps the forward diffusion process subject-independent and injects subject information only into the learned reverse denoising process. The subject embedding therefore does not change the Gaussian corruption schedule; instead, it modulates the trajectory by which the model estimates and removes noise. This separation preserves the stability of standard DDPM training while allowing the reverse process to adapt to subject-dependent signal morphology and artifact patterns. The model comprises three core components:

- (1) a *standard, subject-independent forward diffusion process* that constructs noisy training states from EEG segments;
- (2) a *subject-conditioned reverse denoising process* in which multi-layer embedding injection adapts the denoising trajectory to a subject domain; and
- (3) a *training and transductive adaptation scheme* that optimizes the backbone and source-subject embeddings during training and estimates a new target-subject embedding from unlabeled target EEG at deployment.

A standard DDPM assumes that the training data are drawn from a single distribution and does not explicitly model the distribution differences between subjects, which limits its denoising performance and transferability in practical cross-subject EEG applications. Different subjects differ in the spatial distribution, spectral characteristics, and contamination patterns of their EEG; in multi-subject datasets and mobile-acquisition settings, electrode contact, background noise, and physiological state all vary, so a static, uniformly parameterized denoising strategy is difficult to keep effective. By adding a subject-aware mechanism on top of the DDPM, SADDPM systematically addresses this cross-subject domain shift while retaining the stability of diffusion modeling, yielding more robust denoising.

3.1 Problem Formulation

We consider N source subjects (domains) indexed by $s \in \mathcal{S} = \{1, \dots, N\}$. For subject s , EEG segments $\mathbf{x}_0 \in \mathbb{R}^{C \times L}$ (C channels, L time samples) follow a subject-specific distribution $\mathcal{D}_s$, and the population distribution is the mixture $\mathcal{D} = \sum_{s \in \mathcal{S}} \pi_s \mathcal{D}_s$ with mixing weights $\pi_s \geq 0$ and $\sum_s \pi_s = 1$. In paired denoising experiments, $\mathbf{x}_0$ denotes the artifact-free target signal. In naturally recorded BCI-IV-2a data, where no artifact-free ground truth is available, $\mathbf{x}_0$ denotes the preprocessed EEG segment used as the empirical target of the diffusion objective rather than a physiologically clean

Table 1. Summary of notation.

Symbol	Meaning
$\mathbf{x}_0, \mathbf{x}_t$	target signal; noisy state at diffusion step t
C, L	number of EEG channels; segment length (samples)
T, t	total diffusion steps; step index
$\beta_t, \alpha_t, \bar{\alpha}_t$	noise variance; $\alpha_t=1-\beta_t$; $\bar{\alpha}_t=\prod_{s \leq t} \alpha_s$
$\epsilon, \hat{\epsilon}_{\theta, \phi}$	injected Gaussian noise; combined predicted noise
$\epsilon_{\theta}^c, \epsilon_{\phi}^s$	content-branch and subject-conditioned residual predictors
ω	subject-discrimination classifier head
$s, \mathbf{e}(s) \in \mathbb{R}^d$	subject/domain label; trainable subject embedding
$s^*, \mathbf{e}^*$	target subject; transductively adapted target embedding
$\mathbf{y}, t_0, K$	observed EEG segment; restoration strength; number of samples
λ_e	regularization weight for target-embedding adaptation
$N, \mathcal{D}_s, \pi_s$	number of subjects; subject distribution; mixing weight
$\gamma(\mathbf{e}), \delta(\mathbf{e})$	FiLM scale and shift
R_s, R	per-subject risk; population risk
$\mathcal{L}_r, \mathcal{L}_o, \mathcal{L}_a$	reverse, orthogonality, subject-discrimination losses
$\mathbf{Z}^c, \mathbf{Z}^s$	content / subject feature matrices over a batch

ground truth. A forward diffusion process (Section 3.2) corrupts $\mathbf{x}_0$ into a noisy state $\mathbf{x}_t$ at step $t \in \{1, \dots, T\}$ by injecting Gaussian noise ϵ , and a denoiser $\hat{\epsilon}_{\theta, \phi}$ is trained to predict ϵ , optionally conditioned on a subject embedding $\mathbf{e}(s) \in \mathbb{R}^d$.

For a predictor $\hat{\epsilon}_{\theta, \phi}$, the per-subject and population denoising risks are

$$R_s(\theta, \phi) = \mathbb{E}_{t, \mathbf{x}_0 \sim \mathcal{D}_s, \epsilon} [\|\epsilon - \hat{\epsilon}_{\theta, \phi}(\mathbf{x}_t, t, \cdot)\|^2], \quad R(\theta, \phi) = \sum_{s \in \mathcal{S}} \pi_s R_s(\theta, \phi), \quad (1)$$

with $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$. The conditioning argument “ $\cdot$ ” is empty for a *subject-agnostic* model $\hat{\epsilon}_{\theta, \phi}(\mathbf{x}_t, t)$ and equals $\mathbf{e}(s)$ for the *subject-aware* model $\hat{\epsilon}_{\theta, \phi}(\mathbf{x}_t, t, \mathbf{e}(s))$. Because $\mathcal{D}_s \neq \mathcal{D}_{s'}$ in general (Section 2.2), a model fit on source subjects faces a distribution shift when applied to a target subject s^* . In this paper we use a *transductive subject-adaptation* setting: before denoising a target subject, the model may access unlabeled EEG segments from that subject to estimate a target embedding $\mathbf{e}^*$, but it does not use target task labels or artifact annotations. The quantities of interest are therefore the adapted target risk $R_{s^*}(\theta, \phi, \mathbf{e}^*)$ and the resulting downstream classification accuracy. The goal of SADDPM is to learn the shared parameters (θ, ϕ) and source embeddings $\{\mathbf{e}(s)\}$ that minimize the population risk (1), while enabling rapid unlabeled adaptation of $\mathbf{e}^*$ for a shifted target subject. Table 1 summarizes the notation.

3.2 Forward and Reverse Diffusion Processes

In SADDPM, EEG denoising follows the forward and reverse diffusion principles of the standard DDPM, but subject information is introduced only in the learned reverse denoising process. The model contains a forward diffusion process, which simulates the progressive injection of Gaussian noise, and a reverse denoising process, which simulates the progressive recovery of the signal. The forward process is shared by all subjects; subject-encoding information modulates the reverse network at each denoising step.

Forward diffusion. The forward process gradually transforms a target signal $\mathbf{x}_0$ into noise $\mathbf{x}_T$ that is approximately isotropic Gaussian. It is realized as a sequence of small Gaussian perturbations:

$$q(\mathbf{x}_{1:T} | \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1}), \quad (2)$$

with each step defined as

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad (3)$$

where β_t is the noise-variance hyperparameter at step t , which is increased slowly according to a predefined schedule so that the signal gradually loses its original structure. To simplify sampling, the DDPM allows the intermediate state at an arbitrary step t to be sampled directly from $\mathbf{x}_0$:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{s=1}^t \alpha_s. \quad (4)$$

In SADDPM, the forward process is kept identical to that of the standard DDPM, with no additional modification, in order to preserve the stability and theoretical consistency of the noise-injection process.

Reverse denoising. The reverse process learns a generative trajectory that recovers the target signal from Gaussian noise, modeled as a parameterized Markov chain:

$$p_{\theta, \phi}(\mathbf{x}_{0:T} | \mathbf{e}) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta, \phi}(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{e}), \quad (5)$$

where each reverse conditional is Gaussian and is conditioned on the subject embedding through the learned reverse model:

$$p_{\theta, \phi}(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{e}) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta, \phi}(\mathbf{x}_t, t, \mathbf{e}), \sigma_t^2 \mathbf{I}). \quad (6)$$

Following the standard DDPM, the reverse variance is fixed to the noise schedule ($\sigma_t^2 = \beta_t$). The mean is reparameterized through the combined dual-decoder noise estimate:

$$\boldsymbol{\mu}_{\theta, \phi}(\mathbf{x}_t, t, \mathbf{e}) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\boldsymbol{\epsilon}}_{\theta, \phi}(\mathbf{x}_t, t, \mathbf{e}) \right), \quad (7)$$

where $\hat{\boldsymbol{\epsilon}}_{\theta, \phi}$ is defined in Eq. (11). Unlike a subject-agnostic DDPM, SADDPM conditions the reverse mean through the individual residual branch: at each step t , $\mathbf{e}$ and the current state $\mathbf{x}_t$ determine a subject-specific correction to the subject-invariant content prediction. This conditioning lets the reverse denoising process adapt to subject-dependent signal morphology and artifact characteristics without modifying the forward diffusion schedule.

Trial-wise restoration from an observed EEG segment. The reverse chain in Eq. (5) can generate subject-consistent EEG from Gaussian noise, but the downstream BCI experiment requires a trial-wise denoised version of an observed segment. Given a preprocessed observed EEG window $\mathbf{y}$ and a subject embedding $\mathbf{e}$, SADDPM performs partial noising followed by reverse refinement. Specifically, for a denoising strength $t_0 \in \{1, \dots, T\}$ and K Monte Carlo samples, we initialize

$$\mathbf{x}_{t_0}^{(k)} = \sqrt{\bar{\alpha}_{t_0}} \mathbf{y} + \sqrt{1 - \bar{\alpha}_{t_0}} \boldsymbol{\epsilon}^{(k)}, \quad \boldsymbol{\epsilon}^{(k)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (8)$$

and run the learned reverse process from t_0 to 0 while conditioning on $\mathbf{e}$. The trial-wise denoised output is

$$\widehat{\mathbf{x}}_0 = \frac{1}{K} \sum_{k=1}^K \widehat{\mathbf{x}}_0^{(k)}. \quad (9)$$

The parameter t_0 controls the restoration strength: a larger t_0 allows stronger artifact suppression but increases the risk of over-smoothing or altering task-relevant EEG structure. This inference rule distinguishes trial-wise denoising from unconditional generation and is the procedure used when SADDPM-denoised segments are passed to the downstream classifier.

3.3 Subject Embedding and Conditional Guidance

The subject embedding $\mathbf{e}$ provides subject-domain information that guides the reverse denoising process. Two encoding strategies are possible:

- (1) **One-hot encoding**: each subject corresponds to a unique one-hot vector, suitable when the number of subjects is small.
- (2) **Learnable embedding**: a trainable low-dimensional vector is learned jointly with the main network, automatically capturing latent inter-subject differences.

In practice, we use learnable embeddings for source subjects. The embedding $\mathbf{e}$ is supplied to the subject-conditioned residual path rather than to the subject-invariant content predictor. It is injected at multiple levels of the individual-difference branch to strengthen conditional control. This mechanism enables SADDPM to recover the common structure of EEG through the content branch while tailoring the residual denoising strategy to a subject domain, improving recovery fidelity and cross-subject stability.

Transductive target-subject adaptation. For a target subject s^* not used to fit the shared backbone, a learnable embedding cannot be assumed to exist a priori. We therefore use a transductive adaptation protocol. Given an unlabeled target set $\mathcal{U}_{s^*} = \{\mathbf{y}_i\}_{i=1}^M$, we introduce a new embedding $\mathbf{e}^*$ initialized by the mean source embedding $\bar{\mathbf{e}} = N^{-1} \sum_{s \in \mathcal{S}} \mathbf{e}(s)$. The backbone and source embeddings are frozen, and only $\mathbf{e}^*$ is optimized with the self-supervised diffusion loss

$$\min_{\mathbf{e}^*} \mathbb{E}_{\mathbf{y} \in \mathcal{U}_{s^*}, t, \epsilon} \left[\left\| \epsilon - \left(\epsilon_\theta^c(\mathbf{x}_t, t) + \epsilon_\phi^s(\mathbf{x}_t, t, \mathbf{e}^*) \right) \right\|^2 \right] + \lambda_e \|\mathbf{e}^* - \bar{\mathbf{e}}\|_2^2, \quad (10)$$

where $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{y} + \sqrt{1 - \bar{\alpha}_t} \epsilon$. This adaptation uses the target subject's unlabeled EEG distribution but does not use motor-imagery class labels, clean targets, or artifact annotations. The auxiliary subject-discrimination loss $\mathcal{L}_a$ is not applied during this step, because the target subject is not a source class of the classifier head. The procedure therefore resolves the embedding issue for target subjects while preserving a label-free deployment setting. After adaptation, trial-wise denoising uses $\mathbf{e}^*$ in Eqs. (8)–(9).

3.4 Network Architecture

3.4.1 Backbone Network. SADDPM adopts an improved one-dimensional convolutional U-Net (1D Conv U-Net) to accommodate the temporal nature of EEG, using symmetric encoder and decoder modules for multi-scale feature extraction and reconstruction. The overall architecture is shown in Figure 1.

- **Encoder.** A stack of 1D convolutional blocks, each containing a convolution layer, group normalization, and a nonlinear activation (Swish). After each block the feature length is halved and the number of channels is increased, extracting local temporal features at progressively coarser scales.

- **Middle block.** Several self-attention blocks at the U-Net bottleneck capture global dependencies and strengthen long-range temporal modeling, which is important for cross-channel and cross-segment EEG patterns.
- **Decoder.** Symmetric to the encoder, built from up-sampling (transposed convolution or interpolation followed by convolution) blocks that progressively restore the temporal resolution. Skip connections pass intermediate encoder features directly to the corresponding decoder layers, preserving fine-grained local information and improving reconstruction quality.
- **Time embedding.** At each diffusion step t , a sinusoidal positional encoding embeds the step index and is injected into the convolutional blocks, making the network aware of the current denoising stage so that it can adjust its strategy accordingly.

On top of this shared backbone, SADDPM uses a dual-decoder design for individual-aware modeling (Figure 1). The shared encoder maps the noisy state $\mathbf{x}_t$ to a latent representation, from which two parallel decoders branch: a *denoised-signal* (content) decoder that models the subject-invariant target/content component, and an *individual-difference* decoder that uses the subject embedding to model the subject-conditioned residual component. The reverse-step estimate is obtained by combining the two branches, while a lightweight classifier head on the subject branch encourages the individual-difference representation to encode subject-discriminative signal information. The design is motivated by the observation that EEG distributions differ significantly across subjects: a single unconditional denoiser may either underfit subject-specific artifacts or absorb them into the main content pathway. The dual-decoder structure therefore routes population-level EEG morphology through the content branch and routes subject-dependent corrections through the individual branch, yielding a reverse denoising step that is adaptive without making the whole predictor subject-specific.

Concretely, the denoised-signal (content) branch, parameterized by θ , predicts a subject-invariant noise component $\epsilon_\theta^c(\mathbf{x}_t, t)$, whereas the individual-difference branch, parameterized by ϕ , predicts a subject-conditioned residual component $\epsilon_\phi^s(\mathbf{x}_t, t, \mathbf{e}(s))$. The combined noise estimate used in Eq. (7) is

$$\widehat{\epsilon}_{\theta, \phi}(\mathbf{x}_t, t, \mathbf{e}(s)) = \epsilon_\theta^c(\mathbf{x}_t, t) + \epsilon_\phi^s(\mathbf{x}_t, t, \mathbf{e}(s)). \quad (11)$$

A classifier head ω reads the subject-branch representation and outputs subject logits for the auxiliary subject-discrimination loss. This definition aligns the architecture with the disentanglement objective: the content branch models common EEG structure, while the individual branch carries the subject-conditioned residual needed for subject-specific denoising.

3.4.2 Subject-Aware Conditioning. The subject embedding is fused into the network through conditional injection so that feature processing and generation adapt to each subject. The mechanism has three elements.

Embedding generation. Each source subject label s is mapped to a trainable low-dimensional embedding $\mathbf{e} = \mathbf{e}(s) \in \mathbb{R}^d$, optimized jointly with the backbone parameters; we use s and $\mathbf{e}(s)$ interchangeably as the conditioning argument when the embedding is available. The embeddings are randomly initialized and refined in a data-driven manner, capturing subject-level differences relevant to denoising.

Injection sites. To preserve the content/residual decomposition, the subject embedding is injected into the individual-difference path at multiple levels. Before the convolutional layers of the residual decoder (and the bottleneck features routed to that decoder), a linear transformation maps $\mathbf{e}$ to the

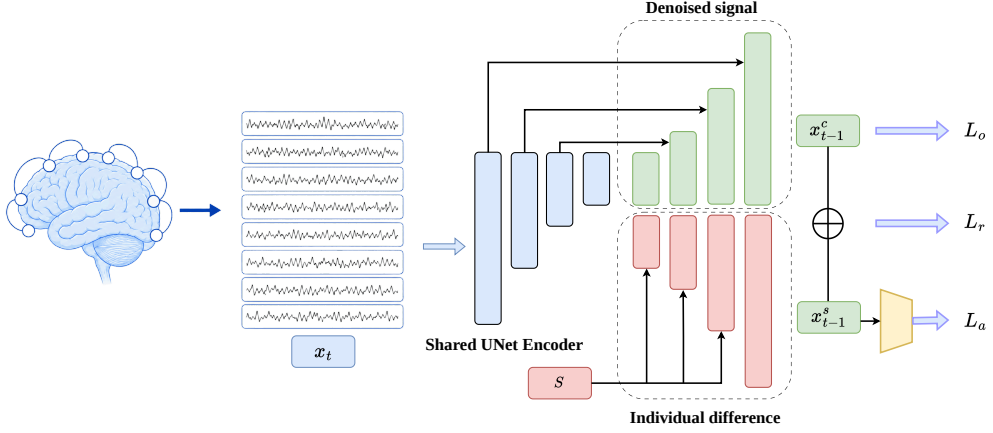


Fig. 1. Overall architecture of SADDPM. Multi-channel EEG is treated as the noisy diffusion state $\mathbf{x}_t$ and passed through a *shared 1D-Conv U-Net encoder*. Two decoder branches share this encoder: a *denoised-signal* (content) decoder that reconstructs the subject-invariant target/content component $\mathbf{x}_{t-1}^c$, and an *individual-difference* decoder that, conditioned on the subject embedding $\mathbf{e}(s)$, models the subject-conditioned residual component $\mathbf{x}_{t-1}^s$. The two branch outputs are combined ($\oplus$) to form the reverse-step estimate $\mathbf{x}_{t-1}$, and a classifier head reads the subject branch. Training uses three losses (Section 3.4.3): the orthogonality loss $\mathcal{L}_o$ on the content branch, the reverse (reconstruction) loss $\mathcal{L}_r$ on the combined estimate, and the angular-margin subject-discrimination loss $\mathcal{L}_a$ on the classifier.

feature-map size, and the result is added to, concatenated with, or used to modulate the residual features.

Fusion via feature modulation. To balance training stability and expressiveness, we use feature-wise modulation (FiLM): scale and shift parameters derived from the subject embedding adjust the output distribution of each convolutional block,

$$\mathbf{h}' = \gamma(\mathbf{e}) \odot \mathbf{h} + \delta(\mathbf{e}), \quad (12)$$

where $\mathbf{h}$ is an intermediate feature, $\odot$ is element-wise multiplication, and $\gamma(\mathbf{e})$ (scale) and $\delta(\mathbf{e})$ (shift) are the modulation parameters produced from the subject embedding by a small multilayer perceptron (MLP). The shift is denoted δ to avoid confusion with the noise-schedule parameter β_t .

3.4.3 Loss Function and Training Objective. SADDPM adopts the same noise-prediction principle as the standard DDPM. For a generic conditional DDPM denoiser ϵ_ψ , the simplified objective can be written as

$$\mathcal{L}_{\text{simple}}(\psi) = \mathbb{E}_{t, \mathbf{x}_0, \epsilon, \mathbf{e}} \left[\left\| \epsilon - \epsilon_\psi \left(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, \mathbf{e} \right) \right\|^2 \right], \quad (13)$$

where $\mathbf{e}$ denotes optional conditioning. In SADDPM, this generic predictor is instantiated by the combined dual-decoder estimate in Eq. (11). At each step, the diffusion index t is sampled uniformly and the noise ϵ from a standard normal; the backbone, branch parameters, and source-subject embeddings are optimized jointly.

To realize the dual-decoder, subject-aware design, the overall training objective combines three terms (Figure 1):

$$\mathcal{L} = \lambda_r \mathcal{L}_r + \lambda_o \mathcal{L}_o + \lambda_a \mathcal{L}_a, \quad (14)$$

where:

- $\mathcal{L}_r$ is the *reverse (reconstruction) loss*, the noise-prediction loss applied to the combined two-branch estimate (Eq. (15)), which drives accurate denoising;
- $\mathcal{L}_o$ is the *orthogonality loss*, which decorrelates the representations of the denoised-content branch and the individual-difference branch, preventing the subject information from being absorbed or ignored by the main branch and keeping the two factors disentangled; and
- $\mathcal{L}_a$ is the *subject-discrimination loss*, an angular-margin (ArcFace-style) classification loss on the subject branch that encourages the individual encoder to be discriminative of subject-level signal structure, using the subject label as supervision.

Concretely, let $\mathbf{Z}^c, \mathbf{Z}^s \in \mathbb{R}^{B \times d}$ stack the column-wise mean-centered content and subject features of a training batch of size B , and let $y \in \{1, \dots, N\}$ be the ground-truth subject of a sample among N subjects (the ℓ_2 /angular normalization in $\mathcal{L}_a$ applies to the classifier features, not to $\mathbf{Z}^c, \mathbf{Z}^s$). The three terms are

$$\begin{aligned} \mathcal{L}_r &= \mathbb{E}_{t, \mathbf{x}_0, \epsilon, s} \left[\left\| \epsilon - \widehat{\epsilon}_{\theta, \phi}(\mathbf{x}_t, t, \mathbf{e}(s)) \right\|^2 \right] \\ &= \mathbb{E}_{t, \mathbf{x}_0, \epsilon, s} \left[\left\| \epsilon - \left(\epsilon_{\theta}^c(\mathbf{x}_t, t) + \epsilon_{\phi}^s(\mathbf{x}_t, t, \mathbf{e}(s)) \right) \right\|^2 \right], \end{aligned} \quad (15)$$

$$\mathcal{L}_o = \left\| \mathbf{Z}^{c\top} \mathbf{Z}^s \right\|_F^2, \quad (16)$$

$$\mathcal{L}_a = -\mathbb{E} \left[\log \frac{e^{\kappa \cos(\vartheta_y + m)}}{e^{\kappa \cos(\vartheta_y + m)} + \sum_{j \neq y} e^{\kappa \cos \vartheta_j}} \right], \quad (17)$$

where $\|\cdot\|_F$ is the Frobenius norm, ϑ_j is the angle between the normalized subject feature and the class- j weight, m is the angular margin, and κ is the ArcFace scale (named κ to avoid clashing with the subject label s). The reconstruction term $\mathcal{L}_r$ corresponds to the $\oplus$ combination of the subject-invariant content prediction and the subject-conditioned residual prediction in Figure 1. If the residual branch outputs zero, the objective reduces to a content-only subject-agnostic DDPM; if the residual branch is active, the subject embedding can correct the common denoising trajectory for subject-specific artifacts. $\mathcal{L}_o$ decorrelates the content and subject representations; and $\mathcal{L}_a$ supervises the classifier head ω . The non-negative weights $\lambda_r, \lambda_o, \lambda_a$ balance the three terms. Using noise prediction as the primary objective stabilizes training and encourages progressive recovery of the target signal across noise levels, while the orthogonality and subject-discrimination terms jointly enhance cross-subject denoising and fidelity.

3.5 Theoretical Analysis

We collect a few short, design-justifying results: *when* subject-conditioning helps (Prop. 3.1); that the subject-aware hypothesis class is at least as expressive as a plain DDPM (Prop. 3.2); that the noise-prediction loss directly controls the reverse-step reconstruction error (Prop. 3.3); what the orthogonality loss enforces (Prop. 3.4); and how the design relates to domain-adaptation theory (Prop. 3.5). These results *motivate* the architecture rather than guarantee performance; proofs are deferred to Appendix A. Unless stated otherwise, expectations are over the data-generating process of Section 3.1, and all quantities are assumed to have finite second moments.

PROPOSITION 3.1 (SUBJECT-CONDITIONING AND DENOISING RISK). *Fix a diffusion step t and consider the squared-error noise-prediction objective of Eq. (13). Let S be the random subject, let $g^*(\mathbf{x}_t, t) = \mathbb{E}[\epsilon \mid \mathbf{x}_t, t]$ be the optimal subject-agnostic predictor with risk $R_u = \mathbb{E}\|\epsilon - g^*\|^2$, and let $h^*(\mathbf{x}_t, t, s) = \mathbb{E}[\epsilon \mid \mathbf{x}_t, t, S=s]$ be the optimal subject-aware predictor with risk $R_c = \mathbb{E}\|\epsilon - h^*\|^2$. Then*

$$R_u - R_c = \mathbb{E}_{\mathbf{x}_t, t} \left[\text{Var}_S(\mathbb{E}[\epsilon \mid \mathbf{x}_t, t, S] \mid \mathbf{x}_t, t) \right] \geq 0. \quad (18)$$

Here $\text{Var}_S(\cdot \mid \mathbf{x}_t, t)$ denotes the total variance (trace of the conditional covariance) of the vector-valued conditional mean.

The gap in (18) is exactly the *between-subject variance* of the optimal denoiser: subject-conditioning lowers the irreducible denoising error precisely to the extent that the optimal denoising map varies across subjects. If all subjects share the same conditional law ($\mathcal{D}_s \equiv \mathcal{D}$), the gap is zero and conditioning cannot help; the larger the inter-subject heterogeneity, the larger the potential gain. This formalizes the motivation for a subject-aware mechanism and is consistent with the empirically uneven gains across subject pairs (Section 4.4).

PROPOSITION 3.2 (THE SUBJECT-AWARE CLASS CONTAINS THE SUBJECT-AGNOSTIC DDPM). *Let $\mathcal{H}_u = \{\epsilon_\theta^c(\mathbf{x}_t, t)\}$ be the subject-agnostic denoiser class and let $\mathcal{H}_c = \{\epsilon_\theta^c(\mathbf{x}_t, t) + \epsilon_\phi^s(\mathbf{x}_t, t, \mathbf{e}(s))\}$ be the dual-decoder FiLM-conditioned subject-aware class of Eq. (11). If the FiLM module can realize the identity modulation $\gamma(\mathbf{e}) \equiv 1$, $\delta(\mathbf{e}) \equiv \mathbf{0}$ and the individual branch can output $\epsilon_\phi^s \equiv \mathbf{0}$, then $\mathcal{H}_u \subseteq \mathcal{H}_c$, and hence for any dataset*

$$\min_{h \in \mathcal{H}_c} \widehat{R}(h) \leq \min_{h \in \mathcal{H}_u} \widehat{R}(h), \quad (19)$$

where $\widehat{R}$ is the empirical (training) risk.

Subject-aware conditioning is thus at least as expressive as a plain DDPM: it can only lower the attainable *training* error. This is a statement about the hypothesis class, not about generalization—the larger class can overfit in the finite-sample regime, which is precisely why we regularize the embeddings (Section 4.3) and flag subject-aware ablations as needed (Section 6).

PROPOSITION 3.3 (NOISE-PREDICTION ERROR CONTROLS THE REVERSE-STEP ERROR). *Let $\widehat{\epsilon}$ be the model's combined noise prediction ($\widehat{\epsilon} = \widehat{\epsilon}_{\theta, \phi}$ in SADDPM) and $\epsilon^\star$ the true noise, with $\|\widehat{\epsilon}(\mathbf{x}_t, t) - \epsilon^\star(\mathbf{x}_t, t)\| \leq \eta_t$. For the DDPM reverse mean $\mu_{\theta, \phi}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\widehat{\epsilon}(\mathbf{x}_t, t))$ and the corresponding $\mu^\star$ obtained by substituting the true noise $\epsilon^\star$, the reverse-step mean error satisfies, at the same $\mathbf{x}_t$,*

$$\|\mu_{\theta, \phi}(\mathbf{x}_t, t) - \mu^\star(\mathbf{x}_t, t)\| = \frac{\beta_t}{\sqrt{\alpha_t} \sqrt{1-\alpha_t}} \|\widehat{\epsilon} - \epsilon^\star\| \leq \frac{\beta_t}{\sqrt{\alpha_t} \sqrt{1-\alpha_t}} \eta_t. \quad (20)$$

Since μ is affine in $\widehat{\epsilon}$, the relation is an exact identity (hence tight): the noise-prediction loss $\mathcal{L}_r$ is not an isolated surrogate but directly governs the per-step reconstruction error, with step gain $c_t = \beta_t/(\sqrt{\alpha_t}\sqrt{1-\alpha_t})$. Summing per-step deviations along a reference trajectory bounds the cumulative mean error by $\sum_t c_t \eta_t$; a fully end-to-end bound would additionally require a Lipschitz/contraction assumption on the reverse kernel, which we do not claim.

PROPOSITION 3.4 (ORTHOGONALITY LOSS ENFORCES DECORRELATION). *Let $\mathbf{Z}^c, \mathbf{Z}^s \in \mathbb{R}^{B \times d}$ be the column-wise mean-centered content and subject feature matrices over a batch of size B , and let $\mathcal{L}_o = \|\mathbf{Z}^{c\top} \mathbf{Z}^s\|_F^2$ as in Eq. (16). Then $\mathcal{L}_o = B^2 \|\widehat{\text{Cov}}(\mathbf{z}^c, \mathbf{z}^s)\|_F^2$, so $\mathcal{L}_o = 0$ if and only if the empirical cross-covariance between the content and subject features vanishes; minimizing $\mathcal{L}_o$ drives the two representations toward (second-order) decorrelation.*

PROPOSITION 3.5 (DOMAIN-ADAPTATION BOUND FOR THE CONTENT REPRESENTATION; RECALLED FROM [4]). *Let D_θ map an input to its content representation, f a downstream hypothesis, and P_S^c, P_T^c the content-representation distributions on a source and a target subject. For binary hypotheses under 0/1 loss,*

$$R_T(f \circ D_\theta) \leq R_S(f \circ D_\theta) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(P_S^c, P_T^c) + \lambda, \quad (21)$$

where $d_{\mathcal{H}\Delta\mathcal{H}}$ is the $\mathcal{H}\Delta\mathcal{H}$ -divergence and λ is the ideal joint risk (the multi-class case is analogous with an appropriate discrepancy).

How the design relates to Eq. (21) (heuristic). SADDPM is *designed* to act on the divergence term: by routing subject-specific variation into the individual branch and decorrelating it from the content branch (Proposition 3.4), the content representation is encouraged toward subject-invariance, which can shrink $d_{\mathcal{H}\Delta\mathcal{H}}(P_S^c, P_T^c)$ and thereby targets cross-subject *adaptation-space drift* [10]. We stress this is *motivation, not a guarantee*: $\mathcal{L}_o$ enforces within-batch content–subject decorrelation, a related but distinct quantity from the cross-domain content divergence in (21), and the term λ is not controlled by our objective.

From ceiling to practice. The benefit quantified in Proposition 3.1 is a population (Bayes-optimal) ceiling: it describes the best achievable benefit of conditioning, not what a finite network with learned embeddings attains. Realizing the gap in (18) requires both embeddings $e(s)$ that are informative about s and sufficient per-subject data to estimate h^* . The modest absolute gains reported in Section 4.4 indicate that only a fraction of this ceiling is realized in practice, which we attribute to limited data per subject and the difficulty of learning fully disentangled embeddings—a gap that the future directions in Section 6 aim to close. Section 5 offers direct empirical support for this mechanism: the subject embedding contributes materially when artifacts are subject-specific—so the between-subject variance in Eq. (18) is non-zero—and becomes nearly inert when artifacts are shared across subjects.

3.6 Conditional Denoiser Variant (SADDPM-Cond)

The SADDPM model described so far is a *generative restoration* denoiser: it learns the target EEG distribution and denoises an observed segment through partial noising followed by subject-conditioned reverse refinement. We additionally study a *conditional* variant, SADDPM-Cond, intended for the paired setting in which (corrupted, clean) examples are available (Section 5). It shares the backbone, subject embedding, and FiLM conditioning of SADDPM, with three changes.

Conditional input. The corrupted observation $\mathbf{y}$ is concatenated with the diffusion state along the channel axis, so the network sees $[\mathbf{x}_t; \mathbf{y}]$ and learns $p(\mathbf{x}_0 | \mathbf{y}, s)$; training uses synthetic pairs (a clean signal plus a known artifact at a controlled signal-to-noise ratio).

$\mathbf{x}_0$ -prediction. Instead of predicting the noise ϵ , SADDPM-Cond predicts the clean signal $\mathbf{x}_0$ directly, which we found markedly more accurate for artifact removal (Section 5).

Full conditional generation. At inference, SADDPM-Cond denoises by *full conditional generation*: the reverse process is run from pure Gaussian noise while conditioning on $\mathbf{y}$ at every step, following the image-to-image diffusion recipe of Palette and SR3 [23, 24]. The output is the denoised estimate $\hat{\mathbf{x}}_0$.

3.7 A Unifying View: Conditioning Reduces Denoising Risk

The benefit of subject conditioning (Proposition 3.1) is one instance of a single principle that also governs the conditional and multi-channel settings of SADDPM-Cond: *conditioning the Bayes-optimal denoiser on additional relevant information reduces its mean-squared error by exactly the corresponding conditional variance*. We record two further instances and a parameterization lemma; proofs are in Appendix A.

PROPOSITION 3.6 (SUBJECT CONDITIONING UNDER AN OBSERVED ARTIFACT). *Consider recovering a clean signal $\mathbf{x}_0$ from a corrupted observation $\mathbf{y}$ (e.g., $\mathbf{y} = \mathbf{x}_0 + \mathbf{a}$ with artifact $\mathbf{a}$), with random subject S . Let the subject-agnostic and subject-aware Bayes denoisers be $\bar{g}(\mathbf{y}) = \mathbb{E}[\mathbf{x}_0 | \mathbf{y}]$ and $\bar{h}(\mathbf{y}, s) = \mathbb{E}[\mathbf{x}_0 | \mathbf{y}, S=s]$, with risks $\bar{R}_u, \bar{R}_c$. Then*

$$\bar{R}_u - \bar{R}_c = \mathbb{E}_{\mathbf{y}}[\text{Var}_S(\mathbb{E}[\mathbf{x}_0 | \mathbf{y}, S] | \mathbf{y})] \geq 0, \quad (22)$$

which is zero iff the Bayes denoiser is subject-invariant ($\bar{h}(\mathbf{y}, S) = \bar{g}(\mathbf{y})$ almost surely)—in particular when $\mathbf{x}_0$ is recoverable from $\mathbf{y}$ alone (a shared artifact)—and strictly positive when the artifact is subject-specific, so that $\mathbf{y}$ under-determines $\mathbf{x}_0$ without S .

This is the theoretical counterpart of the functional-relevance result in Section 5.3: a shared artifact makes the embedding nearly inert ($\Delta\text{CC} \approx 0$), whereas a subject-specific artifact makes the embedding materially useful.

PROPOSITION 3.7 (JOINT VERSUS PER-CHANNEL DENOISING). *For channel c , let the per-channel Bayes denoiser use only that channel’s observation, $\mathbb{E}[\mathbf{x}_{0,c} \mid \mathbf{y}_c]$, and the joint denoiser use all channels, $\mathbb{E}[\mathbf{x}_{0,c} \mid \mathbf{y}_{1:C}]$. Then*

$$\sum_c \mathbb{E} \|\mathbf{x}_{0,c} - \mathbb{E}[\mathbf{x}_{0,c} \mid \mathbf{y}_{1:C}]\|^2 \leq \sum_c \mathbb{E} \|\mathbf{x}_{0,c} - \mathbb{E}[\mathbf{x}_{0,c} \mid \mathbf{y}_c]\|^2, \quad (23)$$

with per-channel gap $\mathbb{E}_{\mathbf{y}_c} [\text{Var}(\mathbb{E}[\mathbf{x}_{0,c} \mid \mathbf{y}_{1:C}] \mid \mathbf{y}_c)]$, which is strictly positive when the artifact has cross-channel (spatial) structure.

This explains why joint multi-channel denoising (Table 7, $\text{CC} \approx 0.99$) exceeds per-channel baselines ($\text{CC} \approx 0.85$): the joint denoiser exploits the artifact’s spatial topography that per-channel models discard.

LEMMA 3.8 (x_0 - AND ϵ -PARAMETERIZATION). *The x_0 - and ϵ -parameterizations of the reverse mean are related by the affine bijection $\mathbf{x}_0 = (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon) / \sqrt{\bar{\alpha}_t}$; hence they share the same Bayes-optimal predictor and differ only in the per-step weighting of the training loss.*

Consequently, the empirical advantage of $\mathbf{x}_0$ -prediction (Table 6, $\text{CC } 0.848 \rightarrow 0.901$) is an optimization and loss-weighting effect—an effective emphasis on low-noise steps, where $\mathbf{x}_0$ is well determined—rather than added model capacity.

4 Experiments

4.1 Dataset

We evaluate the proposed method on the BCI Competition IV-2a dataset, a standard benchmark for multi-class motor-imagery EEG analysis. Its well-structured, multi-subject, multi-session design makes it well suited to assessing cross-subject modeling and the generalization of denoising algorithms.

The dataset contains EEG recordings from nine healthy subjects (A01–A09). Each subject completed two sessions on different days (Session T and Session E), used respectively for training and testing. Each session comprises six runs of 48 trials, giving 288 trials per subject per session. Every trial follows a uniform protocol: the subject first fixates on a central cross (0–2 s); a directional cue (left, right, down, or up, corresponding to imagined movement of the left hand, right hand, both feet, or tongue) then appears for about 1.25 s; the subject performs the corresponding motor imagery until about 6 s into the trial, followed by a short rest. Each trial lasts about 8 s and comprises a preparation, execution, and recovery phase. Such motor imagery elicits event-related desynchronization and synchronization (ERD/ERS) over the sensorimotor cortex [22].

EEG was recorded with 22 channels in a Cz-referenced montage at a 250 Hz sampling rate; three additional EOG channels were recorded to support artifact analysis. All signals were hardware-filtered (0.5–100 Hz) and acquired with high-quality equipment to ensure consistency. We use the data of all nine subjects and adopt the official split without manual resampling or truncation, keeping Session T (training) and Session E (testing) strictly separate to emulate the independent-test scenario of real deployment. Because the dataset covers four motor-imagery classes and contains

natural artifacts (e.g., eye blinks and muscle activity), it provides a solid basis for studying EEG denoising under complex noise.

4.2 Data Preprocessing

We apply standardized preprocessing to improve signal quality and ensure consistency across subjects. For channel selection, we retain all 22 EEG channels and discard the reference and EOG channels to focus on cortical activity. We then apply a 1–50 Hz finite-impulse-response (FIR) band-pass filter to remove low-frequency baseline drift and high-frequency EMG interference while preserving the cognitively and motor-relevant bands, followed by a 50 Hz notch filter to suppress power-line interference and its harmonics. Recordings with a sampling rate above 250 Hz are resampled to 250 Hz to keep the input dimension consistent and reduce computational load. We segment the continuous EEG with a sliding window of 2 s and a step of 0.5 s, generating a sufficient number of training samples while maintaining coverage. Finally, to remove distribution shifts caused by differing amplitude scales across subjects, each segment is normalized per channel to zero mean and unit variance, which stabilizes training and improves cross-subject generalization.

4.3 Experimental Setup

To ensure stable learning under complex noise and across subjects, SADDPM uses a unified, structured set of hyperparameters. We optimize with Adam at an initial learning rate of 1×10^{-4} , combined with a cosine-annealing schedule for smooth decay and stable late-stage convergence. The batch size is 64, and the diffusion process uses $T = 1000$ steps with a linear schedule for the noise level β_t , ensuring a gradual transition from the noisy state to the empirical target representation. Training runs for 100 epochs over the source Session T set. For source-domain individual modeling, the source subject is assigned a 128-dimensional learnable embedding that captures subject-dependent signal and artifact characteristics; this embedding is optimized jointly with the backbone and injected at multiple stages of the reverse process. To improve generalization and avoid overfitting the embeddings to the training data, we apply L_2 regularization with weight 1×10^{-4} to the embedding vectors. When the evaluation target differs from the source, the target embedding is not assumed to exist beforehand; it is adapted from unlabeled target-subject EEG windows as described below. All experiments run on the same hardware (an NVIDIA A100 GPU), with identical training configurations across methods to ensure fair comparison. The overall source-training procedure is summarized in Algorithm 1.

To assess cross-subject transfer, we adopt a pairwise source–target protocol with transductive target-subject adaptation. For each row of the matrix, the shared SADDPM backbone and the source embedding are trained only on the Session T data of one source subject. For each target column, before denoising Session E, we freeze the backbone and optimize a target embedding from unlabeled target Session T EEG windows using Eq. (10); no target motor-imagery labels, clean targets, or artifact annotations are used. In this single-source setting, the source-embedding mean used to initialize the adapted target embedding reduces to the learned source embedding. The adapted embedding is then used in the trial-wise partial-noising restoration procedure of Eqs. (8)–(9). The restoration strength t_0 and the number of reverse samples K are fixed across all source–target pairs.

The downstream four-class motor-imagery classifier is trained on denoised source Session T trials and evaluated on denoised target Session E trials; target labels are used only to compute the final classification accuracy. The resulting 9×9 accuracy matrix has diagonal entries for session-separated within-subject transfer and off-diagonal entries for cross-subject transfer. For each target (column) subject, the row labeled “Mean” reports the average accuracy over all source subjects, summarizing how well denoised representations transfer to that target. This protocol is

Algorithm 1 SADDPM source training.

Require: Source training set of preprocessed EEG segments with subject labels $\{(\mathbf{x}_0, s)\} \sim q(\mathbf{x}_0, s)$; noise schedule $\{\beta_t\}_{t=1}^T$

- 1: Initialize the source embedding(s); if more than one source subject is used, pre-train the subject classifier ω on source labels
- 2: **while** not converged **do**
- 3: Sample a source training segment and its subject label $(\mathbf{x}_0, s) \sim q(\mathbf{x}_0, s)$
- 4: Sample a diffusion step $t \sim \text{Uniform}(\{1, \dots, T\})$
- 5: Sample noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 6: Form the noisy state $\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \epsilon$
- 7: Compute $\epsilon_\theta^c(\mathbf{x}_t, t)$, $\epsilon_\phi^s(\mathbf{x}_t, t, \mathbf{e}(s))$, $\hat{\epsilon}_{\theta, \phi}$, and the subject logits.
- 8: Compute $\mathcal{L} = \lambda_r \mathcal{L}_r + \lambda_o \mathcal{L}_o + \lambda_a \mathcal{L}_a$, with $\lambda_a = 0$ in single-source training
- 9: Update θ, ϕ, ω and the source embedding(s) by a gradient-descent step on $\mathcal{L}$
- 10: **end while**

Table 2. Downstream classification accuracy (%) after ICA denoising. Rows are the source (training) subject; columns are the target (test) subject. The diagonal is the within-subject case; “Mean” is the average over source subjects for each target.

Source ↓ / Target →	s1	s2	s3	s4	s5	s6	s7	s8	s9
s1	89.29	39.29	25.93	50.00	40.74	27.27	62.96	29.63	36.00
s2	82.61	92.86	33.33	41.67	33.33	31.82	33.33	37.04	36.00
s3	53.57	85.71	81.48	58.33	33.33	54.55	44.44	33.33	48.00
s4	53.57	46.43	81.48	91.67	33.33	54.55	37.04	37.04	44.00
s5	42.86	35.71	37.04	91.67	88.89	54.55	40.74	33.33	40.00
s6	50.00	39.29	37.04	50.00	77.78	90.91	44.44	33.33	44.00
s7	57.14	50.00	37.04	41.67	55.56	90.91	74.07	33.33	40.00
s8	39.29	39.29	44.44	50.00	59.26	45.45	66.67	74.07	44.00
s9	32.14	32.14	37.04	45.83	44.44	45.45	33.33	74.07	84.00
Mean	55.61	51.19	46.09	57.87	51.85	55.05	48.56	42.80	46.22

transductive rather than fully zero-shot: it evaluates whether SADDPM can adapt to a deployment subject’s unlabeled signal distribution while preserving a label-free target setting.

4.4 Results and Analysis

To evaluate the effectiveness of SADDPM for denoising motor-imagery EEG, we conduct systematic experiments on BCI Competition IV-2a and compare against the classical ICA baseline. The evaluation has two aspects: (i) the effect of denoising on downstream classification, and (ii) the ability of SADDPM to preserve subject-specific signal structure.

4.4.1 Downstream Classification. Tables 2 and 3 report the downstream classification accuracy (%) of the cross-subject train–test combinations after ICA and SADDPM denoising, respectively. For each combination, the denoised 2 s segments are fed to a downstream four-class motor-imagery classifier, and we report its classification accuracy; the same classifier and training protocol are applied to both ICA- and SADDPM-denoised data to ensure a fair comparison.

Overall, SADDPM attains more stable cross-subject classification than ICA. Averaging the per-target means (the “Mean” row), SADDPM reaches 52.83% versus 50.58% for ICA, and it improves the mean accuracy for eight of the nine target subjects, with s4 the only exception and the gain on s6

Table 3. Downstream classification accuracy (%) after SADDPM denoising. Layout as in Table 2.

Source ↓ / Target →	s1	s2	s3	s4	s5	s6	s7	s8	s9
s1	85.59	46.43	46.43	52.00	37.04	31.82	68.22	37.04	38.46
s2	85.71	90.01	34.28	44.00	44.44	37.27	32.14	44.18	35.77
s3	60.71	92.86	82.11	58.00	58.15	55.00	49.29	51.85	44.62
s4	52.86	47.21	78.57	88.45	48.15	50.00	38.57	40.74	46.15
s5	45.86	35.71	37.04	91.67	85.47	52.00	42.81	36.12	46.15
s6	52.14	35.00	46.43	44.44	79.12	88.00	49.15	33.33	50.00
s7	55.71	50.00	42.16	52.00	55.56	90.91	76.12	43.10	47.28
s8	39.29	39.29	44.44	47.25	59.26	45.45	70.37	78.88	44.00
s9	25.15	35.71	32.14	38.20	44.44	45.45	34.07	73.08	85.15
Mean	55.89	52.47	49.29	57.33	56.85	55.10	51.19	48.70	48.62

Table 4. Aggregate downstream classification summary. “All” averages all 81 source–target entries; “off-diagonal” excludes the within-subject entries and therefore isolates cross-subject transfer; “diagonal” averages the 9 within-subject entries.

Metric	ICA	SADDPM	Δ
All source–target pairs	50.58	52.83	+2.25
Off-diagonal cross-subject pairs	46.25	48.88	+2.63
Diagonal within-subject pairs	85.25	84.42	−0.83
Std. of per-target means	4.76	3.35	−1.41

marginal. As summarized in Table 4, the improvement is concentrated in off-diagonal source–target pairs: SADDPM improves the cross-subject average from 46.25% to 48.88% (+2.63 points), whereas the within-subject diagonal average decreases slightly from 85.25% to 84.42% (−0.83 points). This pattern is consistent with the goal of improving cross-subject robustness rather than optimizing the session-separated within-subject case.

The improvement is most evident when the source subject has transferable signal structure. With s3 as the source subject, for instance, SADDPM raises accuracy on several targets substantially (e.g., s5: 33.33% → 58.15% and s8: 33.33% → 51.85%), indicating that its denoised representations transfer well. By contrast, when s9 is the source subject, both methods perform poorly across targets, suggesting a higher proportion of non-additive artifacts or spectral aliasing in s9; in this case neither method holds a clear advantage.

The absolute gains are modest, but SADDPM exhibits smaller fluctuations across target subjects: the standard deviation of the per-target mean accuracy is 3.35 for SADDPM versus 4.76 for ICA. For target subjects with large distribution gaps (e.g., s8 and s9 as the test subject), the transductively adapted subject embedding provides a mechanism for adjusting the reverse denoising trajectory to the target signal distribution. These results suggest that subject-aware diffusion can stabilize downstream representations under cross-subject distribution shift. Because BCI-IV-2a does not provide artifact-free ground truth, however, this phase should be interpreted as indirect evidence based on downstream decoding rather than a definitive artifact-level denoising evaluation.

4.4.2 Subject-Consistency Analysis. To examine how faithfully SADDPM-generated signals reflect real signals, we compute subject-correlation matrices (Table 5): panel (a) reports correlations within the original BCI-IV signals, and panel (b) reports correlations between SADDPM-generated samples and the real signals. Each entry is the average correlation between the (per-channel, trial-averaged)

Table 5. Subject-correlation analysis. (a) Within-subject and cross-subject correlation coefficients of the original BCI-IV EEG. (b) Correlation coefficients between SADDPM-generated samples and the real BCI-IV EEG. In both panels, the diagonal (within-subject) coefficients are generally the largest in their rows, suggesting retention of subject-specific signal structure.

(a) Original BCI-IV EEG										
Subj. ↓ / →	s1	s2	s3	s4	s5	s6	s7	s8	s9	
s1	0.102	0.047	0.055	0.059	0.045	0.051	0.050	0.042	0.043	
s2	0.048	0.074	0.040	0.038	0.033	0.041	0.045	0.037	0.038	
s3	0.056	0.040	0.080	0.048	0.037	0.043	0.048	0.037	0.039	
s4	0.059	0.038	0.048	0.088	0.041	0.042	0.054	0.043	0.044	
s5	0.045	0.033	0.037	0.041	0.058	0.034	0.039	0.031	0.034	
s6	0.053	0.041	0.043	0.042	0.033	0.071	0.042	0.034	0.038	
s7	0.050	0.046	0.048	0.041	0.039	0.043	0.092	0.042	0.050	
s8	0.042	0.037	0.036	0.053	0.031	0.054	0.042	0.069	0.032	
s9	0.050	0.038	0.038	0.043	0.034	0.038	0.043	0.032	0.066	

(b) SADDPM-generated samples vs. real BCI-IV EEG										
Gen. ↓ / Real →	s1	s2	s3	s4	s5	s6	s7	s8	s9	
s1	0.109	0.071	0.077	0.080	0.066	0.073	0.072	0.063	0.070	
s2	0.073	0.090	0.067	0.065	0.059	0.066	0.068	0.056	0.061	
s3	0.074	0.065	0.096	0.072	0.060	0.069	0.070	0.060	0.063	
s4	0.077	0.063	0.073	0.100	0.063	0.068	0.074	0.064	0.064	
s5	0.068	0.060	0.065	0.067	0.076	0.062	0.063	0.055	0.059	
s6	0.083	0.066	0.069	0.067	0.060	0.088	0.068	0.057	0.061	
s7	0.072	0.069	0.079	0.075	0.071	0.069	0.103	0.066	0.066	
s8	0.066	0.064	0.065	0.069	0.058	0.062	0.066	0.082	0.058	
s9	0.071	0.064	0.067	0.067	0.060	0.065	0.067	0.056	0.082	

signals of the two corresponding subjects, averaged over channels, so that the diagonal measures within-subject self-similarity and the off-diagonal entries measure cross-subject similarity.

In the SADDPM-generated samples, the within-subject correlations are generally higher than the cross-subject correlations, matching the structure of the real EEG. For example, subjects s1, s4, and s7 have within-subject coefficients of 0.109, 0.100, and 0.103, respectively, all larger than their cross-subject values. This suggests that SADDPM retains subject-specific signal morphology while learning the global EEG distribution. We interpret this matrix as a subject-consistency check rather than evidence of biometric identity preservation or complete artifact-level fidelity: correlation alone cannot establish that all physiological task-relevant components are preserved. The result nevertheless supports the use of subject-aware conditioning for personalized EEG preprocessing and downstream decoding.

5 Paired Ground-Truth Denoising

The study so far evaluates denoising *indirectly*, through downstream classification, because BCI-IV-2a provides no artifact-free ground truth. To measure denoising fidelity *directly*, we additionally adopt the paired protocol of EEGdenoiseNet [31]: a clean EEG segment is combined with a known artifact—ocular (EOG) or myogenic (EMG)—at a controlled signal-to-noise ratio (SNR), so the recovered signal is scored against the true clean signal. We report the correlation coefficient with the clean signal (CC $\uparrow$), averaged over the SNR range -7 to 2 dB. These experiments use the conditional variant SADDPM-Cond (Section 3.6); we study a single-channel benchmark and a multi-channel setting in which the EEGdenoiseNet artifacts are injected into the 22-channel

Table 6. Single-channel denoising on the EEGdenoiseNet benchmark (CC $\uparrow$ with the clean signal, averaged over -7 to 2 dB). CNN baselines are from EEGdenoiseNet [31].

Method	EOG CC	EMG CC
Noisy (input)	0.500	0.506
SimpleCNN	0.912	0.742
ComplexCNN	0.909	0.691
NovelCNN	0.878	0.831
SADDPM-Cond (ours)	0.901	0.680

windows with a physiological spatial topography (ocular frontal, myogenic lateral), which also provides subject labels for the subject-ablation study.

Setup. SADDPM-Cond is trained on synthetic pairs: a clean EEG segment is combined with an EEGdenoiseNet artifact at an SNR drawn uniformly over -7 to 2 dB (a low-SNR regime, -16 to -6 dB, is used where noted), and the network learns to recover the clean segment. The optimizer and schedule follow Section 4.3 (Adam, 1×10^{-4} , cosine, batch 64, $T = 1000$, 128-dimensional subject embedding with L_2 weight decay 1×10^{-4}). In the multi-channel setting, artifacts are injected into all 22 channels with a physiological spatial topography and the denoiser is trained jointly over channels; pairs are split by trial into disjoint train and test sets. For the subject ablation, the “wrong” embedding $\mathbf{e}(s')$ is that of a different, randomly chosen subject, and the “null” embedding is the dedicated no-subject slot; inference uses full conditional generation throughout.

The decisive design choice. For the conditional denoiser, predicting the clean signal directly ($\mathbf{x}_0$ -prediction) rather than the noise is the main lever: on the single-channel EOG benchmark it raises CC from 0.848 to 0.901. Combined with full conditional generation (Section 3.6), this makes SADDPM-Cond competitive with supervised CNN denoisers on ocular-artifact removal.

5.1 Single-Channel Benchmark

On the standard single-channel EEGdenoiseNet benchmark (Table 6, Figure 2), SADDPM-Cond is *competitive* with supervised CNN denoisers on ocular artifacts (EOG CC 0.901 vs. SimpleCNN 0.912 and ComplexCNN 0.909) and *trails* CNN baselines more clearly on the harder broadband myogenic artifact (EMG CC 0.680 vs. SimpleCNN 0.742 and the best reported baseline, NovelCNN, 0.831). We therefore claim viability, not superiority: a diffusion denoiser can approach standard supervised CNNs on EOG, while EMG denoising remains more challenging. Example denoised traces (Figure 3) show the recovered signal closely following the clean ground truth while the ocular drift is suppressed.

5.2 Multi-Channel Joint Denoising

Denoising all 22 channels jointly lets SADDPM-Cond exploit the artifact’s spatial topography (Proposition 3.7), reaching CC ≈ 0.99 for both artifact types (Table 7, Figure 4), well above per-channel CNN baselines, which operate on one channel at a time by construction.

5.3 Subject-Aware Conditioning Is Functionally Relevant

To test whether the subject embedding contributes materially, we denoise each segment three ways—with the *correct* embedding $\mathbf{e}(s)$, a *wrong* embedding $\mathbf{e}(s')$, and a *null* embedding—and measure the drop Δ CC between correct and wrong (Table 8, Figure 5). When the artifact is *shared* across subjects, the embedding has negligible effect (Δ CC ≈ 0): the clean signal is fully recoverable

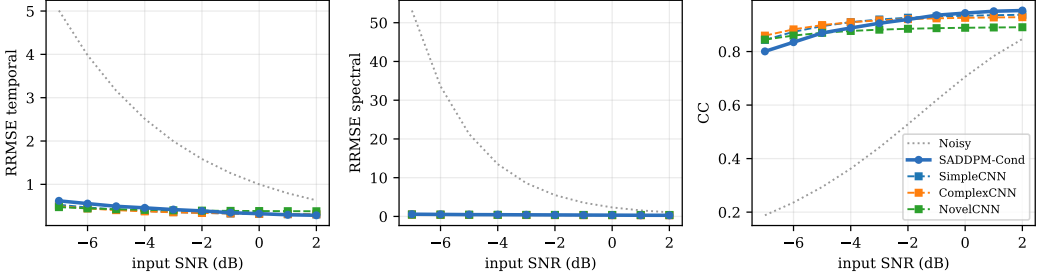


Fig. 2. Single-channel denoising on the EEGdenoiseNet benchmark (EOG). $RRMSE_{\text{temporal}}$, $RRMSE_{\text{spectral}}$, and correlation coefficient (CC) versus input SNR for SADDPM-Cond and the EEGdenoiseNet CNN base-lines; “Noisy” is the unprocessed input. SADDPM-Cond tracks the supervised CNNs across the SNR range, establishing diffusion as a viable EEG denoiser.

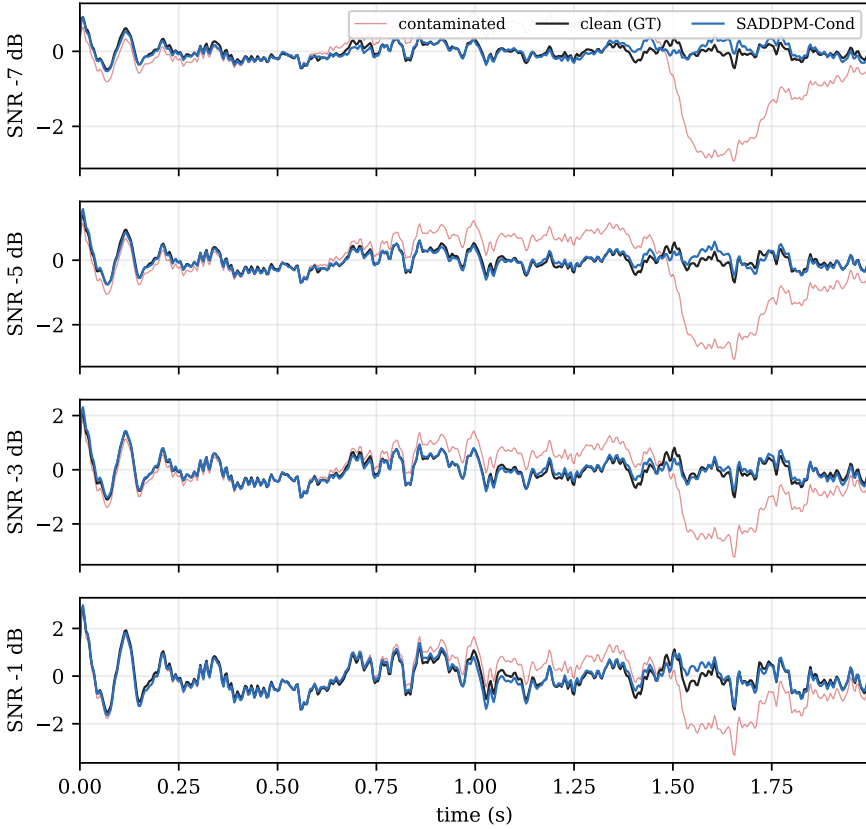


Fig. 3. Example EOG-contaminated segments at four input SNRs. The SADDPM-Cond-denoised signal (blue) recovers the clean ground truth (black) and suppresses the ocular drift present in the input (red).

from the input, so the subject index provides no additional predictive information. When each subject’s artifact occupies a *distinct spatial subspace* (physiologically plausible, reflecting anatomy

Table 7. Multi-channel joint denoising (CC $\uparrow$), identical test windows. The per-channel SimpleCNN is applied independently to each channel.

Method	EOG CC	EMG CC
Noisy (input)	0.662	0.542
Per-channel SimpleCNN [31]	0.844	0.853
SADDDPM-Cond (multi-channel, ours)	0.988	0.994

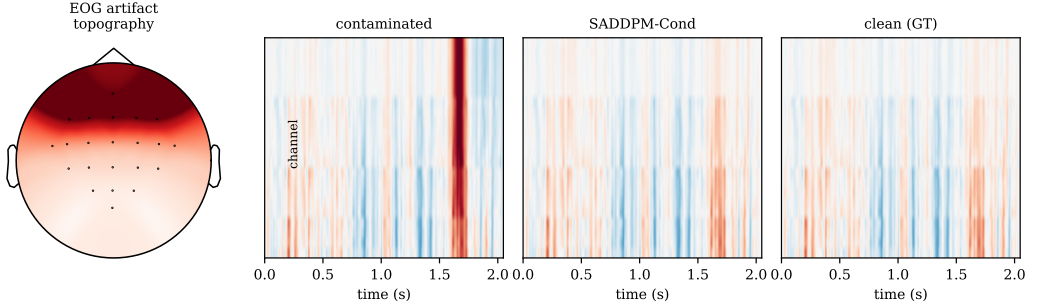


Fig. 4. Multi-channel joint denoising (EOG). Left: the frontal-dominant spatial topography of the injected ocular artifact. Right: contaminated, SADDDPM-Cond-denoised, and clean 22-channel windows (channel $\times$ time); the spatially structured artifact is removed and the denoised output matches the ground truth.

Table 8. Subject ablation for SADDDPM-Cond (EOG, CC $\uparrow$): denoising with the correct vs. a wrong vs. a null subject embedding. Δ CC is correct–wrong; Δ CC ≥ 0.02 indicates a functionally relevant embedding.

Regime	correct $e(s)$	wrong $e(s')$	null	Δ CC
Shared artifact	0.988	0.987	0.982	+0.001
Subject-specific artifact	0.993	0.828	0.865	+0.165
+ low SNR	0.972	0.699	0.791	+0.273

and electrode differences), the embedding becomes functionally relevant—a wrong embedding removes the wrong subspace and CC collapses from 0.993 to 0.828 (Δ CC = +0.165), rising to +0.273 at low SNR. The effect is robust across artifact-specificity strength (Δ CC = +0.080/+0.165/+0.139 at topography gain 0.3/0.6/1.0) and holds for EMG (+0.087).

This is the empirical counterpart of Proposition 3.6: the value of subject conditioning equals the between-subject variance of the optimal denoiser, which is zero for shared artifacts and large for subject-specific ones. We therefore state the subject-aware claim with its condition: conditioning is functionally relevant *when artifacts are subject-specific*, which is physiologically the typical case.

Scope. These results establish *viability*, not superiority. The multi-channel and subject-specific settings use semi-synthetic data (artifacts injected with modeled topographies), as is standard for EEGdenoiseNet-style paired evaluation; the subject-specific topography is a physiologically-motivated modeling choice. We do not claim that diffusion is the best learned denoiser, nor that subject-awareness uniquely requires diffusion; the contribution is that a subject-aware diffusion denoiser is effective and competitive, and that its subject mechanism is functionally relevant under subject-specific artifacts.

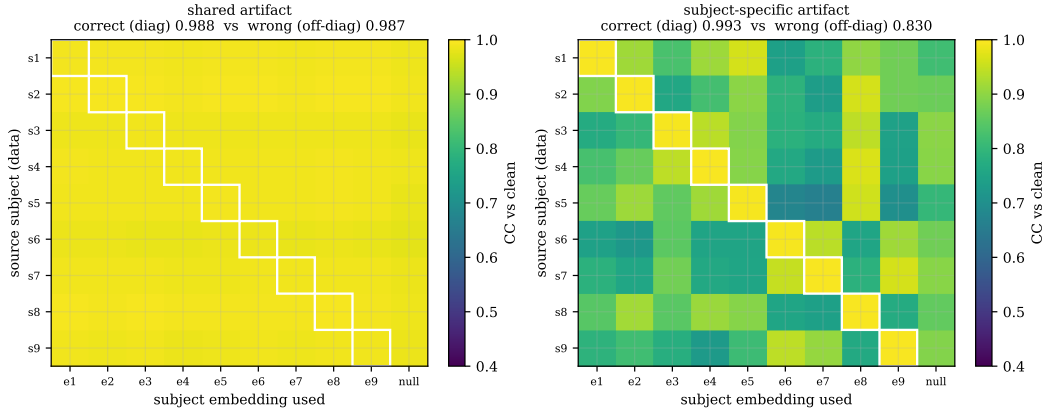


Fig. 5. Is the subject embedding functionally relevant? Each cell is the denoising CC when subject i 's data is denoised with subject j 's embedding (diagonal = correct subject). With a shared artifact (left) the embedding is irrelevant (correct $\approx$ wrong); with subject-specific artifacts (right) only the correct subject's embedding recovers the signal—a bright diagonal against darker off-diagonal entries—showing that subject conditioning contributes materially when artifacts are subject-specific (cf. Table 8 and Proposition 3.6).

6 Conclusion and future work

This work tackles EEG denoising under the cross-subject generalization challenge that limits multi-subject deep-learning pipelines. We designed and implemented SADDPM, a framework that couples a denoising diffusion probabilistic model with a subject-aware mechanism: trainable low-dimensional subject embeddings are injected as conditioning signals at multiple network levels, so the model adapts its reverse denoising trajectory to each individual while retaining the training stability of diffusion. For target subjects without a pre-trained embedding, a transductive, label-free step estimates only the embedding from unlabeled target-domain EEG, keeping the shared backbone fixed.

Across two complementary protocols on BCI Competition IV-2a, the approach proves viable. Under downstream motor-imagery classification, SADDPM is more stable than the classical ICA baseline, with a mean-accuracy gain concentrated in cross-subject transfer and generated signals that preserve subject-specific structure. Under the EEGdenoiseNet paired ground-truth protocol, the conditional variant SADDPM-Cond is competitive with supervised CNN denoisers on ocular artifacts, attains high-fidelity multi-channel denoising ($CC \approx 0.99$), and exhibits subject conditioning that is functionally relevant precisely when artifacts are subject-specific. We read these as evidence of feasibility rather than superiority over the strongest learned denoisers.

Current subject adaptation still relies on unlabeled target data; reducing this dependence through transfer, few-shot, or zero-shot strategies—together with stronger baselines, more datasets and artifact types, end-to-end joint denoising-decoding, and faster sampling for low-latency BCI—are natural next steps. More broadly, SADDPM offers a novel and practical route to subject-adaptive generative denoising for multi-subject EEG, extends diffusion models to individual-aware neural time-series processing, and frames EEG denoising as an autonomous and adaptive systems problem: a learning-based BCI component that must stay reliable under changing users and cross-user distribution shift [5].

Ethics and Privacy Statement

This study uses only the publicly available, de-identified BCI Competition IV-2a dataset; no new human-subjects data were collected, and no personally identifying information is used. EEG signals are sensitive physiological data, and models trained on them could in principle be misused for individual profiling; the subject-aware component here is intended solely to improve cross-subject denoising for legitimate analysis and brain-computer interface research, and the learned subject embeddings are not designed for biometric identification of individuals. We see no additional societal risks beyond those already considered by the original dataset’s collection and release.

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A Proofs

For completeness we first recall two standard facts about DDPMs (which are not contributions of this work) and then prove the propositions (and the parameterization lemma) of Sections 3.5 and 3.7.

LEMMA A.1 (CLOSED-FORM FORWARD MARGINAL [14]). *With $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, the forward process of Eqs. (2)–(3) satisfies $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$.*

PROOF. Write the single-step update as $\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1}$ with $\boldsymbol{\epsilon}_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ i.i.d. Unrolling one step gives

$$\mathbf{x}_t = \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t-1})} \boldsymbol{\epsilon}_{t-2} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1}.$$

The last two terms are independent zero-mean Gaussians with variances $\alpha_t(1 - \alpha_{t-1})$ and $1 - \alpha_t$, which sum to $1 - \alpha_t \alpha_{t-1}$; hence they merge into a single $\mathcal{N}(\mathbf{0}, (1 - \alpha_t \alpha_{t-1})\mathbf{I})$ term. Iterating down to $\mathbf{x}_0$ yields mean $(\prod_{s=1}^t \sqrt{\alpha_s}) \mathbf{x}_0 = \sqrt{\bar{\alpha}_t} \mathbf{x}_0$ and covariance $(1 - \prod_{s=1}^t \alpha_s)\mathbf{I} = (1 - \bar{\alpha}_t)\mathbf{I}$. $\square$

Simplified training objective. Minimizing the variational bound on $-\log p_\theta(\mathbf{x}_0)$ reduces, after the loss reweighting of Ho et al. [14], to the noise-prediction objective $\mathcal{L}_{\text{simple}}$ of Eq. (13), which we use throughout; see [14] for the full derivation.

PROOF OF PROPOSITION 3.1. Fix t and abbreviate $g^\star = \mathbb{E}[\boldsymbol{\epsilon} | \mathbf{x}_t, t]$ and $h^\star = \mathbb{E}[\boldsymbol{\epsilon} | \mathbf{x}_t, t, S]$. By the tower property of conditional expectation, $g^\star = \mathbb{E}[h^\star | \mathbf{x}_t, t]$. Decompose the subject-agnostic risk as

$$R_u = \mathbb{E} \|\boldsymbol{\epsilon} - g^\star\|^2 = \mathbb{E} \|\boldsymbol{\epsilon} - h^\star\|^2 + \mathbb{E} \|h^\star - g^\star\|^2 + 2 \mathbb{E} \langle \boldsymbol{\epsilon} - h^\star, h^\star - g^\star \rangle.$$

Conditioning the cross term on $(\mathbf{x}_t, t, S)$ and using $\mathbb{E}[\boldsymbol{\epsilon} - h^\star | \mathbf{x}_t, t, S] = \mathbf{0}$, while $h^\star - g^\star$ is $(\mathbf{x}_t, t, S)$ -measurable, shows that the cross term vanishes. Hence $R_u = R_c + \mathbb{E} \|h^\star - g^\star\|^2$. Finally, since $g^\star = \mathbb{E}[h^\star | \mathbf{x}_t, t]$,

$$\mathbb{E} \|h^\star - g^\star\|^2 = \mathbb{E}_{\mathbf{x}_t, t} [\text{Var}_S(h^\star | \mathbf{x}_t, t)] \geq 0,$$

which is (18). $\square$

PROOF OF PROPOSITION 3.2. Setting $\boldsymbol{\gamma}(\mathbf{e}) \equiv \mathbf{1}$ and $\boldsymbol{\delta}(\mathbf{e}) \equiv \mathbf{0}$ makes every FiLM layer (Eq. (12)) the identity map $\mathbf{h}' = \mathbf{h}$, so the conditioning input $\mathbf{e}(s)$ has no effect and $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{e}(s)) = \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$; additionally taking $\boldsymbol{\epsilon}_\phi \equiv \mathbf{0}$ makes the dual-decoder output coincide with the subject-agnostic denoiser. Hence every element of $\mathcal{H}_u$ is realized by some element of $\mathcal{H}_c$, i.e. $\mathcal{H}_u \subseteq \mathcal{H}_c$. The minimum of $\widehat{R}$ over a superset is no larger than over the subset, which gives (19). (This is a statement about representational capacity and empirical risk; it does not bound the generalization gap.) $\square$

PROOF OF PROPOSITION 3.3. Both $\boldsymbol{\mu}_\theta$ and $\boldsymbol{\mu}^\star$ use the same $\mathbf{x}_t$ and differ only through the noise term, which enters affinely. Subtracting,

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) - \boldsymbol{\mu}^\star(\mathbf{x}_t, t) = -\frac{1}{\sqrt{\alpha_t}} \cdot \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} (\widehat{\boldsymbol{\epsilon}}(\mathbf{x}_t, t) - \boldsymbol{\epsilon}^\star(\mathbf{x}_t, t)),$$

since the $\mathbf{x}_t$ terms cancel. Taking norms gives the equality in (20), and applying $\|\widehat{\boldsymbol{\epsilon}} - \boldsymbol{\epsilon}^\star\| \leq \eta_t$ gives the bound. The coefficient $c_t = \beta_t / (\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t})$ is a fixed positive scalar, so the relation is tight. $\square$

PROOF OF PROPOSITION 3.4. Because the columns of $\mathbf{Z}^c$ and $\mathbf{Z}^s$ are mean-centered, the empirical cross-covariance of the two feature sets is $\widehat{\text{Cov}}(\mathbf{z}^c, \mathbf{z}^s) = \frac{1}{B} \mathbf{Z}^{c\top} \mathbf{Z}^s$. Therefore

$$\mathcal{L}_o = \|\mathbf{Z}^{c\top} \mathbf{Z}^s\|_F^2 = B^2 \|\widehat{\text{Cov}}(\mathbf{z}^c, \mathbf{z}^s)\|_F^2 \geq 0,$$

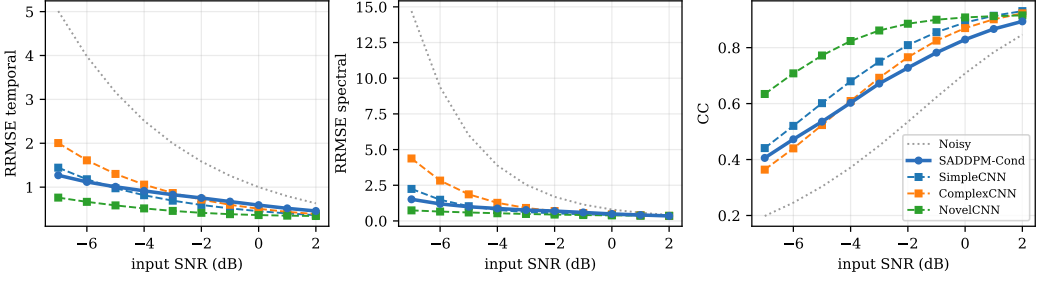


Fig. 6. Single-channel denoising on the EEGdenoiseNet benchmark (EMG; cf. Figure 2). For the harder broadband myogenic artifact, SADDPM-Cond trails the supervised CNNs (mean CC 0.680 vs. SimpleCNN 0.742).

which equals zero if and only if every entry of the cross-covariance matrix is zero, i.e., the content and subject features are pairwise uncorrelated. Minimizing $\mathcal{L}_o$ thus enforces second-order decorrelation; note that decorrelation does not in general imply statistical independence. $\square$

PROOF OF PROPOSITION 3.5. This is the bound of Ben-David et al. [4] (their Theorem 2), applied with the source and target distributions taken to be the content-representation distributions P_S^c, P_T^c induced by D_θ ; we restate it here only for reference and claim no novelty. The multi-class extension follows analogously with a discrepancy-style divergence in place of $d_{\mathcal{H}\Delta\mathcal{H}}$. $\square$

PROOF OF PROPOSITION 3.6. The argument mirrors that of Proposition 3.1 with y in place of $(\mathbf{x}_t, t)$. Abbreviate $\bar{g} = \mathbb{E}[\mathbf{x}_0 \mid y]$ and $\bar{h} = \mathbb{E}[\mathbf{x}_0 \mid y, S]$; by the tower property $\bar{g} = \mathbb{E}[\bar{h} \mid y]$. Expanding $\bar{R}_u = \mathbb{E}\|\mathbf{x}_0 - \bar{g}\|^2$ and noting that the cross term $\mathbb{E}\langle \mathbf{x}_0 - \bar{h}, \bar{h} - \bar{g} \rangle$ vanishes (condition on (y, S) : $\mathbb{E}[\mathbf{x}_0 - \bar{h} \mid y, S] = \mathbf{0}$ while $\bar{h} - \bar{g}$ is (y, S) -measurable) gives $\bar{R}_u = \bar{R}_c + \mathbb{E}\|\bar{h} - \bar{g}\|^2$, and $\mathbb{E}\|\bar{h} - \bar{g}\|^2 = \mathbb{E}_y[\text{Var}_S(\bar{h} \mid y)]$, which is (22). The gap is zero iff $\bar{h}$ is y -measurable, i.e. $\bar{h}(y, S) = \bar{g}(y)$ a.s.; if $\mathbf{x}_0 = f(y)$ for some measurable f (shared artifact) then $\bar{h} = \bar{g} = f$ and the gap vanishes, whereas a subject-specific artifact makes $\bar{h}$ depend on S and the gap positive. $\square$

PROOF OF PROPOSITION 3.7. Since $\mathbf{y}_c$ is a sub-vector of $\mathbf{y}_{1:C}$, the conditional expectation given the larger collection is the L^2 projection onto a larger subspace, so for each c , $\mathbb{E}\|\mathbf{x}_{0,c} - \mathbb{E}[\mathbf{x}_{0,c} \mid \mathbf{y}_{1:C}]\|^2 \leq \mathbb{E}\|\mathbf{x}_{0,c} - \mathbb{E}[\mathbf{x}_{0,c} \mid \mathbf{y}_c]\|^2$, with gap $\mathbb{E}_{\mathbf{y}_c}[\text{Var}(\mathbb{E}[\mathbf{x}_{0,c} \mid \mathbf{y}_{1:C}] \mid \mathbf{y}_c)]$ by the same tower-property decomposition as Proposition 3.1. Summing over c gives (23); the gap is strictly positive whenever the other channels are informative about $\mathbf{x}_{0,c}$ given $\mathbf{y}_c$, i.e. when the artifact has cross-channel structure. $\square$

PROOF OF LEMMA 3.8. For fixed $(\mathbf{x}_t, t)$ the map $\epsilon \mapsto \mathbf{x}_0 = (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon) / \sqrt{\bar{\alpha}_t}$ is an affine bijection, so a predictor of one quantity induces a predictor of the other with the same conditional expectation; the Bayes-optimal solutions coincide. Substituting the relation into the squared-error objective rescales the per-step loss by $(1 - \bar{\alpha}_t) / \bar{\alpha}_t$, so the two parameterizations optimize the same risk under different per-timestep weights. $\square$

B Additional Results: EMG Artifacts and Spectral Recovery

Figures 6–10 provide the myogenic (EMG) counterparts of the main-text ocular (EOG) results, together with the spectral-recovery view. EMG is a broadband artifact and is harder than EOG, as the single-channel benchmark (Figure 6) shows.

Received 06 Month 2026; revised XX Month 2026; accepted XX Month 2026

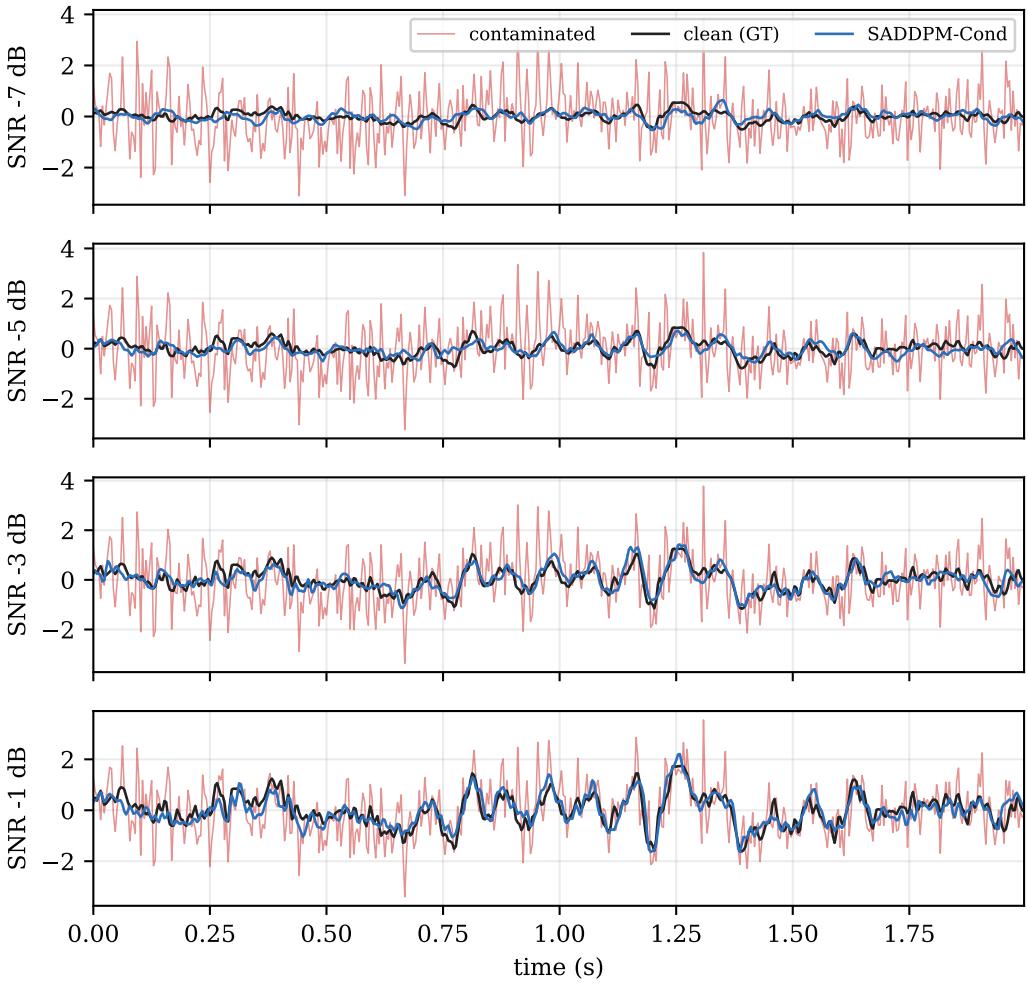


Fig. 7. Example EMG-contaminated segments at four input SNRs (cf. Figure 3).

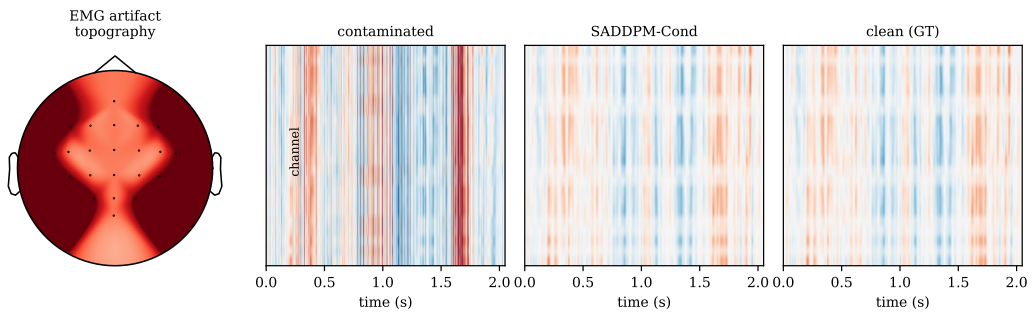


Fig. 8. Multi-channel joint denoising for the laterally distributed myogenic artifact (cf. Figure 4); joint denoising reaches CC ≈ 0.99 .

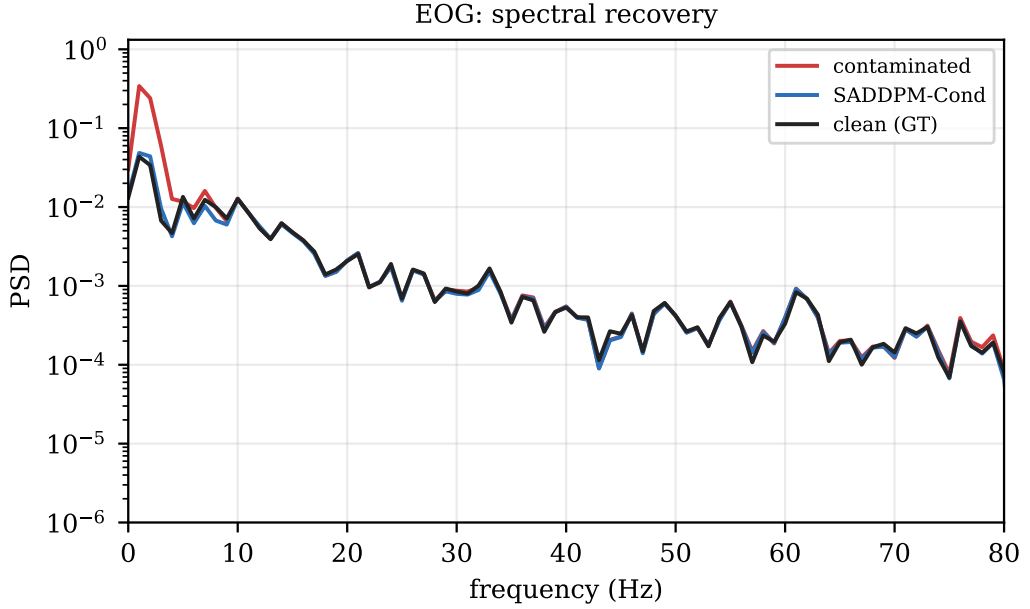


Fig. 9. Spectral recovery (EOG): the SADDPM-Cond-denoised power spectral density (blue) matches the clean spectrum (black), removing the excess low-frequency power introduced by the ocular artifact (red).

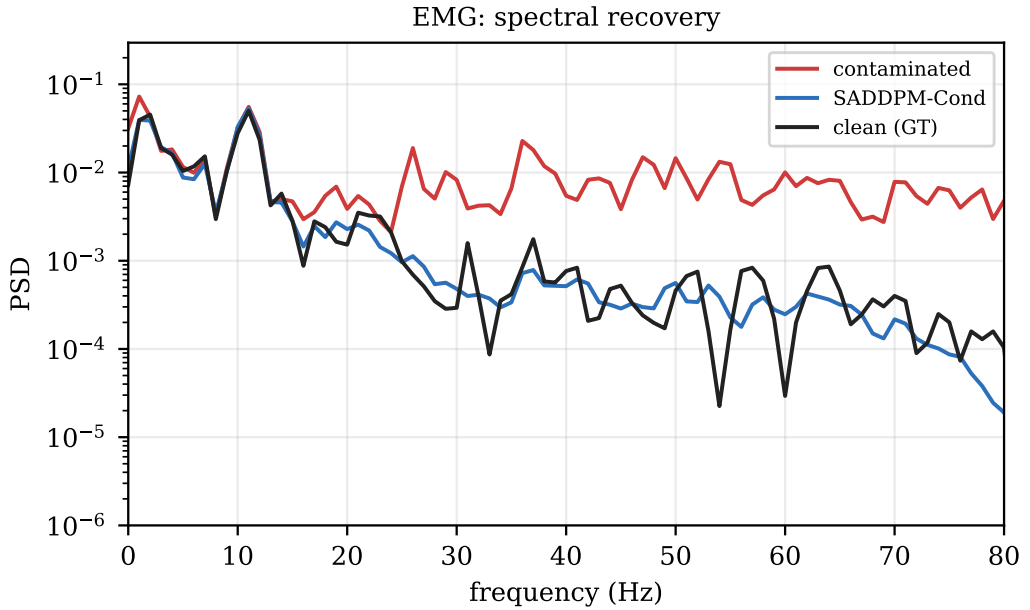


Fig. 10. Spectral recovery (EMG; cf. Figure 9).